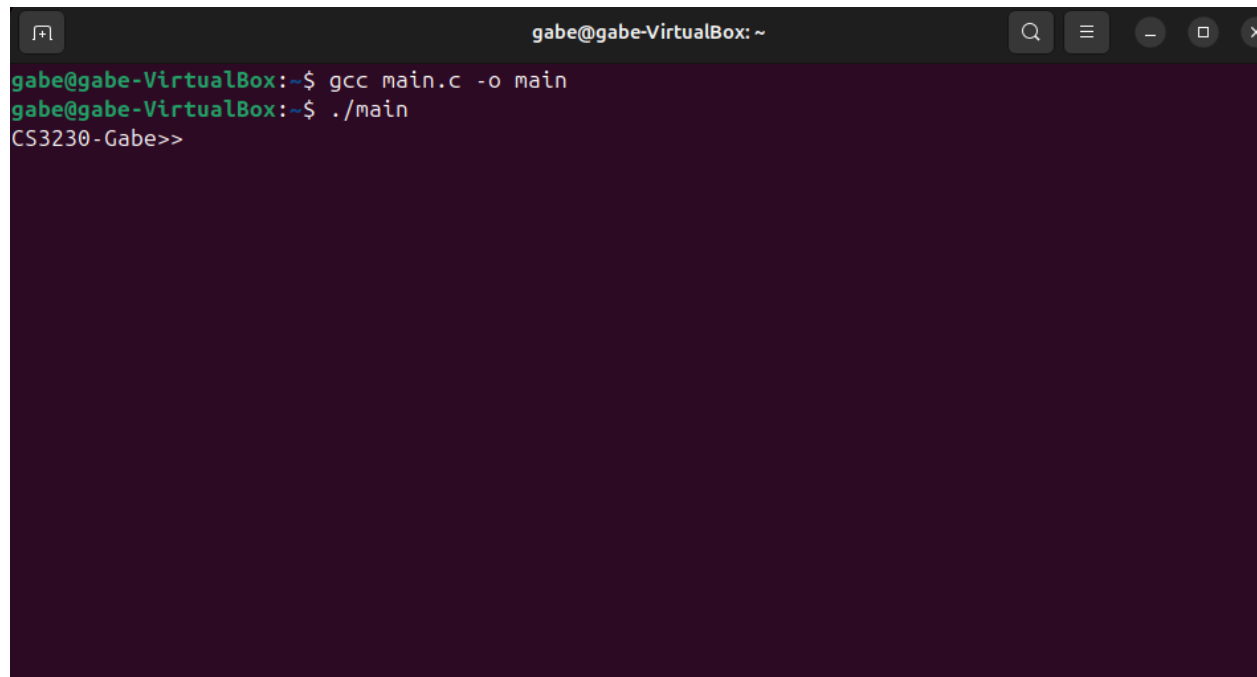
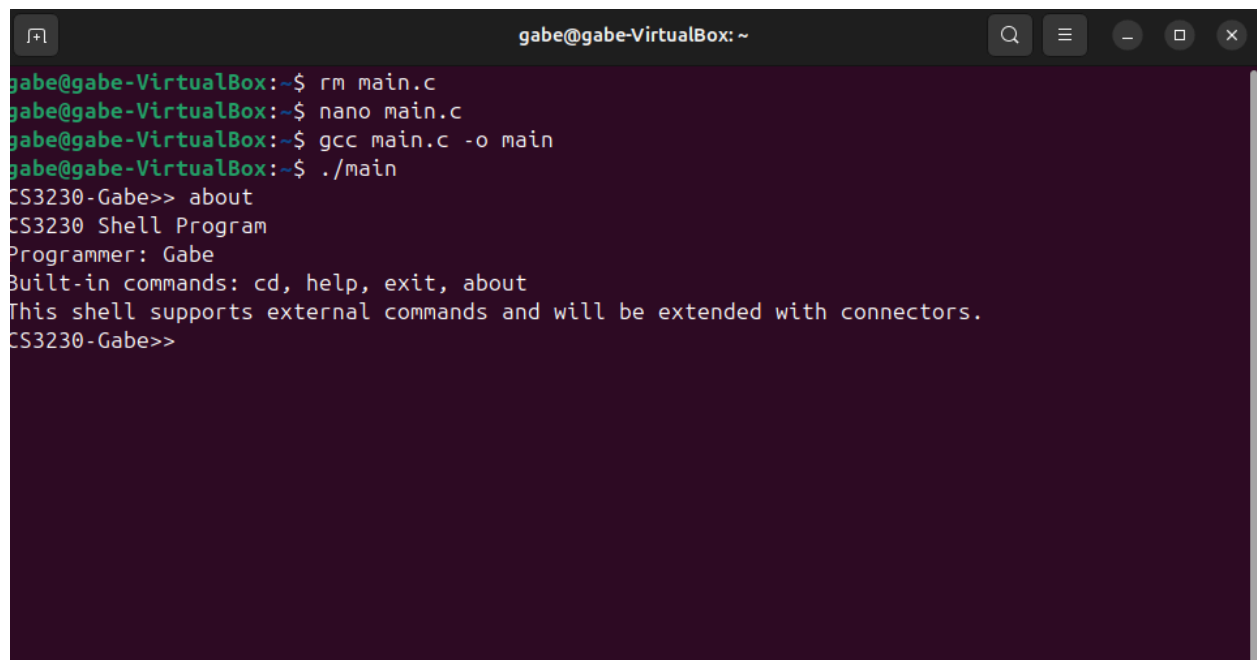


Compile the shell:

A terminal window titled 'gabe@gabe-VirtualBox: ~' with search, menu, and window control icons in the title bar. The terminal shows the following commands and output:

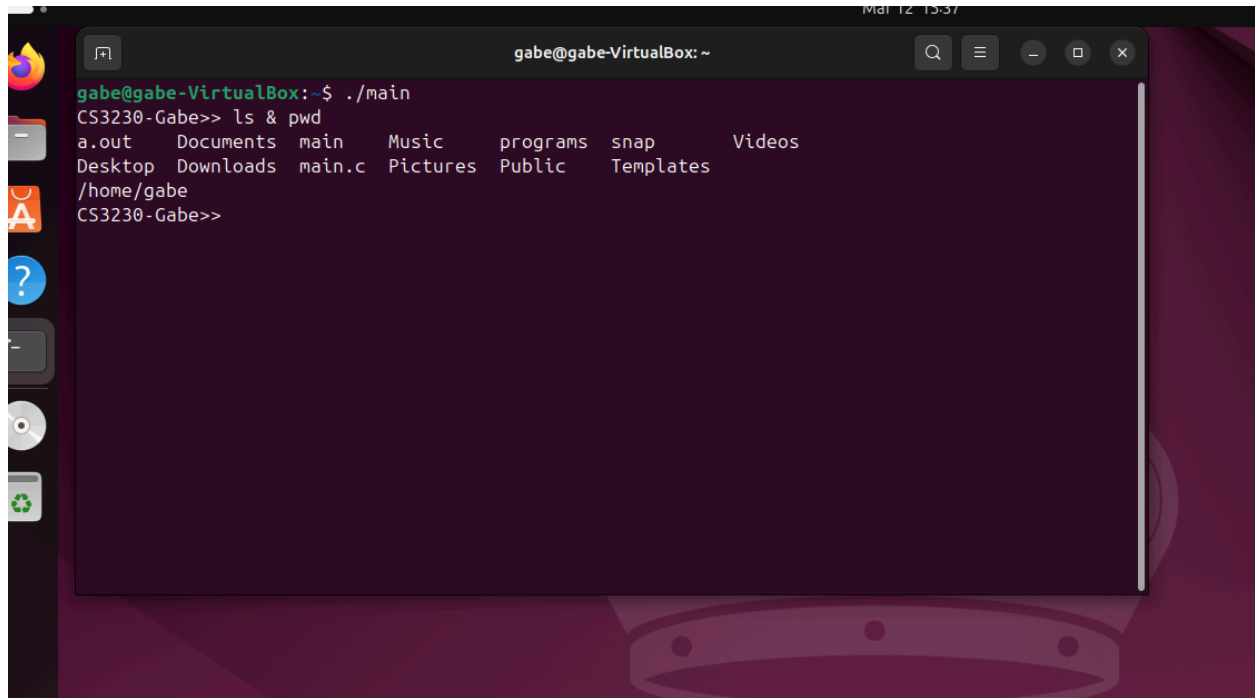
```
gabe@gabe-VirtualBox:~$ gcc main.c -o main
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>>
```

Run basic command examples:

A terminal window titled 'gabe@gabe-VirtualBox: ~' with search, menu, and window control icons in the title bar. The terminal shows the following commands and output:

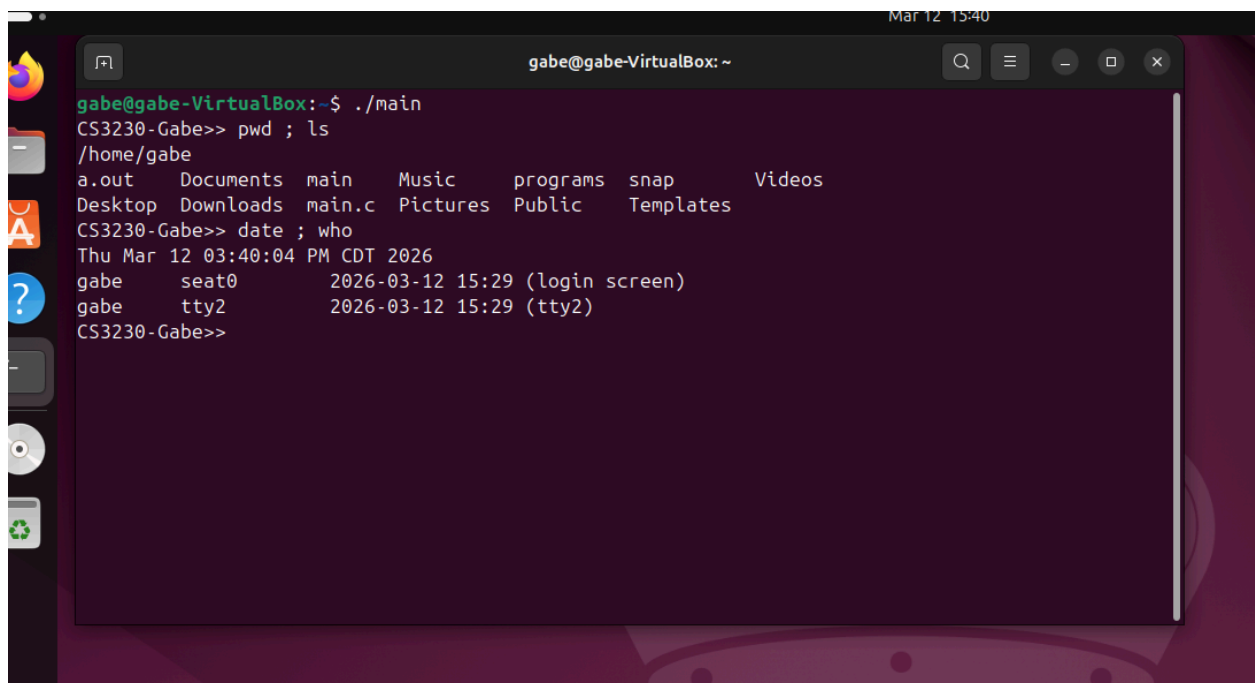
```
gabe@gabe-VirtualBox:~$ rm main.c
gabe@gabe-VirtualBox:~$ nano main.c
gabe@gabe-VirtualBox:~$ gcc main.c -o main
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>> about
CS3230 Shell Program
Programmer: Gabe
Built-in commands: cd, help, exit, about
This shell supports external commands and will be extended with connectors.
CS3230-Gabe>>
```

Run ls & pwd command:

A terminal window titled 'gabe@gabe-VirtualBox: ~' with a search icon, menu icon, and window controls. The prompt is 'gabe@gabe-VirtualBox:~\$./main'. The output shows 'CS3230-Gabe>> ls & pwd' followed by a directory listing: 'a.out Documents main Music programs snap Videos' and 'Desktop Downloads main.c Pictures Public Templates' on the next line. The path '/home/gabe' is shown, and the prompt returns to 'CS3230-Gabe>>'.

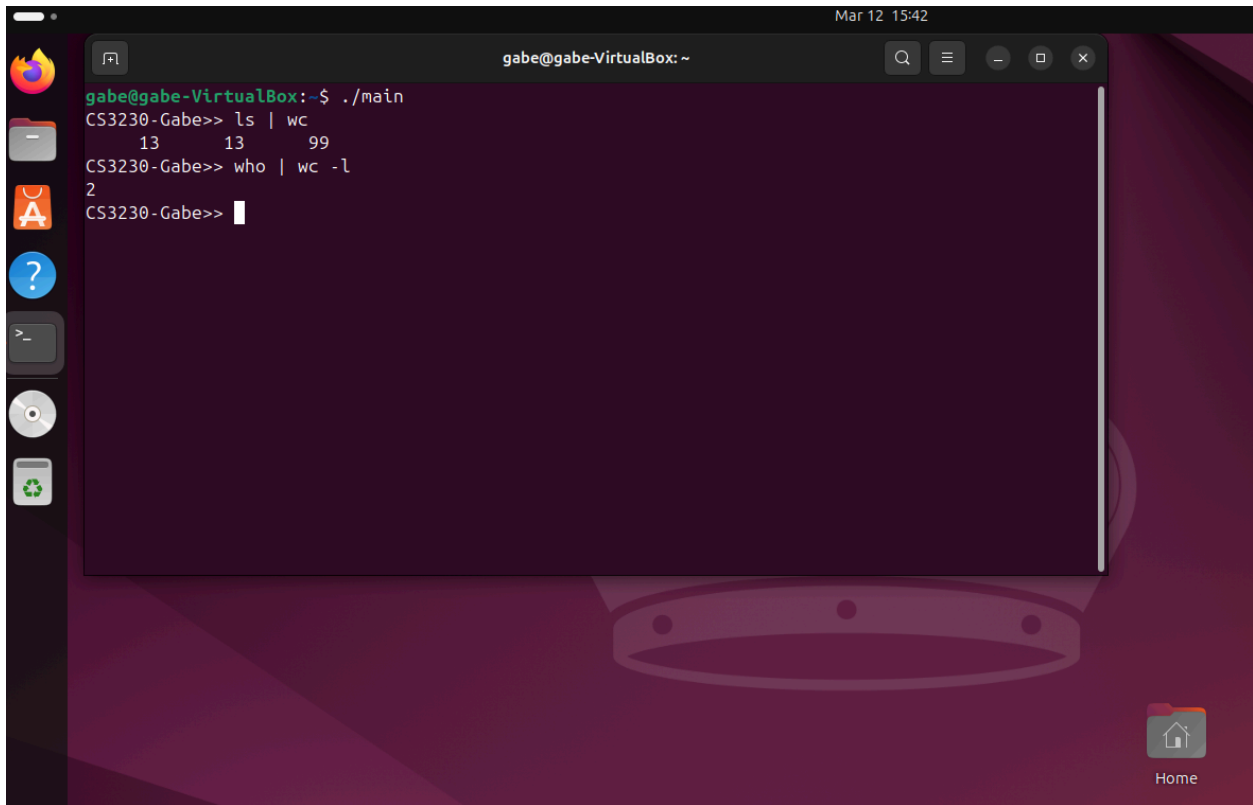
```
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>> ls & pwd
a.out  Documents  main    Music    programs  snap      Videos
Desktop Downloads  main.c  Pictures Public    Templates
/home/gabe
CS3230-Gabe>>
```

Run the date and who commands:

A terminal window titled 'gabe@gabe-VirtualBox: ~' with a search icon, menu icon, and window controls. The prompt is 'gabe@gabe-VirtualBox:~\$./main'. The output shows 'CS3230-Gabe>> pwd ; ls' followed by the path '/home/gabe' and a directory listing: 'a.out Documents main Music programs snap Videos' and 'Desktop Downloads main.c Pictures Public Templates'. Then, 'CS3230-Gabe>> date ; who' is executed, showing the date 'Thu Mar 12 03:40:04 PM CDT 2026' and user information: 'gabe seat0 2026-03-12 15:29 (login screen)' and 'gabe tty2 2026-03-12 15:29 (tty2)'. The prompt returns to 'CS3230-Gabe>>'.

```
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>> pwd ; ls
/home/gabe
a.out  Documents  main    Music    programs  snap      Videos
Desktop Downloads  main.c  Pictures Public    Templates
CS3230-Gabe>> date ; who
Thu Mar 12 03:40:04 PM CDT 2026
gabe    seat0      2026-03-12 15:29 (login screen)
gabe    tty2       2026-03-12 15:29 (tty2)
CS3230-Gabe>>
```

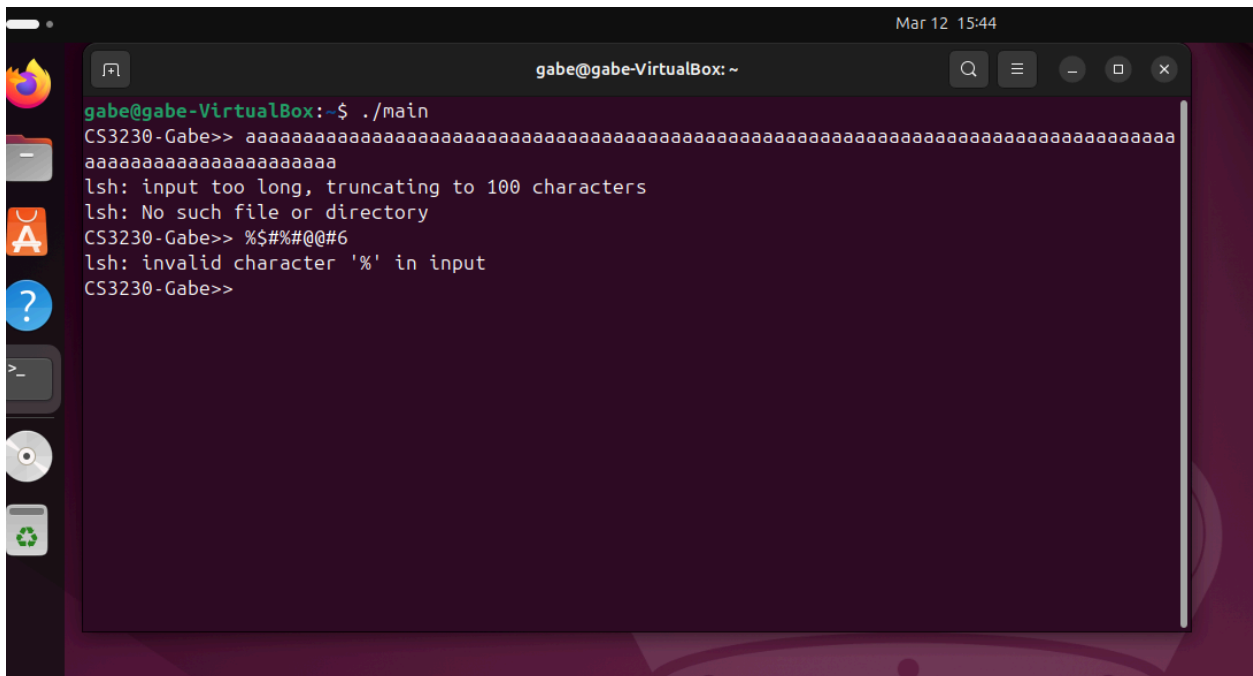
Pipe command demonstration:



A terminal window titled 'gabe@gabe-VirtualBox: ~' with a search, menu, and window control bar. The terminal shows the execution of a program named 'main'. The user runs 'ls | wc' and receives output showing 13 files, 13 lines, and 99 characters. Then, the user runs 'who | wc -l' and receives output showing 2 lines. The terminal is set against a dark purple background with a faint image of a device.

```
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>> ls | wc
      13      13      99
CS3230-Gabe>> who | wc -l
      2
CS3230-Gabe>>
```

Check for invalid characters:



The same terminal window as above, but now testing for invalid characters. The user enters a long string of 'a's, which is truncated to 100 characters. Then, the user enters '%\$##@#6', which results in an error message: 'lsh: invalid character '%' in input'.

```
gabe@gabe-VirtualBox:~$ ./main
CS3230-Gabe>> aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
aaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
lsh: input too long, truncating to 100 characters
lsh: No such file or directory
CS3230-Gabe>> %$##@#6
lsh: invalid character '%' in input
CS3230-Gabe>>
```